

Sufficient and Necessary Conditions for the Parameters of Conditional Entropies

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This standalone document compiles Sections 4 and 5 of the main manuscript as a concise, human-readable extract. It gives the sufficient and necessary parameter conditions for the conditional entropies introduced in Ref. [4], covering general measures, both tropical endpoints, and derivations without assuming finiteness of the singular weighted moment at $\alpha = 1$. The proof sketches retain the central ideas—finite-support factorization, common discretization, and localized high-dimensional counterexamples—while omitting the long analytic calculations. Complete analytic proofs are given in Ref. [6], and their formal counterparts are documented in Ref. [5].

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4. SUFFICIENT CONDITIONS FOR THE PARAMETERS

In this section, we establish sufficient conditions for the parameters under which the homomorphisms and derivations (and, consequently, the associated conditional entropies) are monotone under conditional majorization. In the following section, we show that these conditions are also necessary, without imposing a separate regularity assumption on the measure.

Throughout Sections 4 and 5, every argument headed “Proof sketch” is completed analytically in Ref. [6]; the corresponding formal statements and their dependencies are documented in Ref. [5]. The sketches below record the mathematical mechanism and the parameter restriction, while those references supply the omitted estimates and formal verification.

a. Basic terminology and notation. A *register* is one component of the joint system: X is the first register and Y is the conditioning register. We write P_{XY} for their joint probability mass function on finite alphabets, $\mathcal{Y}$ for the alphabet of Y , and

$$P_Y(y) := \sum_x P_{XY}(x, y), \quad P_{X|Y=y}(x) := \frac{P_{XY}(x, y)}{P_Y(y)}$$

when $P_Y(y) > 0$. A zero-mass conditional distribution may be chosen arbitrarily because every formula below either omits it or multiplies it by $P_Y(y) = 0$. A *column* is the nonnegative vector $P_{XY}(\cdot, y)$.

A *conditionally mixing channel* stochastically processes Y and, in each resulting branch, applies a doubly stochastic mixing map to the X -column; doubly stochastic means that the map is nonnegative and every row and column sums to one. Conditional majorization is the preorder generated by these channels. A preorder is a reflexive and transitive comparison, and *monotone* means order-preserving for this comparison. The corresponding semiring consists of finite nonnegative joint matrices with the direct-sum and tensor-product operations of Ref. [4]. A *homomorphism* preserves those operations and their units. A *derivation* is additive and obeys the product rule relative to a chosen homomorphism.

The general forms of the homomorphisms and derivations in Ref. [4] depend on a finite measure: in the temperate case one may write it as $\mu = t\tau$, with τ a probability measure and its sign encoded by t . However, not all such choices are valid, as some lead to violations of monotonicity under conditionally mixing channels. While monotonicity under mixing channels acting on the first register always holds, monotonicity under channels acting on the conditioning register fails for some choices of the parameters. We therefore examine the convexity properties of the homomorphism as a function of the joint probability matrix. Proposition A.1 and Lemma A.3 of Ref. [4] identify these convexity properties with the required monotonicity and lift the columnwise result to the original semiring and channel preorder of Eqs. (43) and (49) therein.

The one-column functional for which we need to establish convexity or concavity is obtained from Eq. (67) and Proposition 3.6 of Ref. [4]:

$$\Phi(P_{XY}) = \Phi_{t,\tau}(P_{XY}) = \sum_{y \in \mathcal{Y}} P_Y(y) \exp \left(t \int_{[0,+\infty]} H_\alpha(P_{X|Y=y}) d\tau(\alpha) \right), \quad (1)$$

where τ is a probability measure and $t \in \mathbb{R} \setminus \{0\}$. Thus it suffices to study, for $\vec{x} \in \mathbb{R}_{\geq 0}^d \setminus \{0\}$,

$$\vec{x} \mapsto \|\vec{x}\|_1 \exp \left(t \int_{[0,+\infty]} H_\alpha(\hat{x}) d\tau(\alpha) \right), \quad \hat{x} = \frac{\vec{x}}{\|\vec{x}\|_1}. \quad (2)$$

Here d is the number of coordinates, $\|\vec{x}\|_1 := \sum_{i=1}^d x_i$, and $\hat{x}$ is the normalized probability vector obtained from $\vec{x}$. For a probability vector $s = (s_i)_i$, the Rényi entropy is

$$H_\alpha(s) := \frac{1}{1-\alpha} \log \left(\sum_i s_i^\alpha \right), \quad \alpha \in (0, +\infty) \setminus \{1\},$$

with endpoint values

$$H_0(s) = \log |\{i : s_i > 0\}|, \quad H_1(s) = - \sum_i s_i \log s_i, \quad H_\infty(s) = - \log \max_i s_i,$$

where $0 \log 0 := 0$. We use these endpoint conventions throughout. The *support* of a vector is its set of nonzero coordinates. For a positive measure ν , its closed support $\text{supp}(\nu)$ is the set of points every neighbourhood of which has positive ν -measure. A point a is an *atom* when $\nu(\{a\}) > 0$, and δ_a is the Dirac probability measure concentrated at a . A discrete, finitely supported measure is one concentrated on finitely many parameter values.

The word *temperate* means that t is finite and nonzero. The positive and negative *tropical* regimes are the limits $t \rightarrow +\infty$ and $t \rightarrow -\infty$, where log-sum-exp becomes a maximum and a

minimum, respectively. A derivation is the first-order $t \rightarrow 0$ object. A *common discretization* uses the same finitely supported approximation of the parameter measure at every vector in one convexity or concavity inequality. For reference, F is convex when

$$F(\lambda x + (1 - \lambda)z) \leq \lambda F(x) + (1 - \lambda)F(z), \quad 0 \leq \lambda \leq 1,$$

and concave when this inequality is reversed. We begin with finite support, pass to a general measure by a common discretization, and finally take the tropical and derivation limits.

A. Sufficient conditions for the discrete case

The discrete case exploits two basic facts. The first is Minkowski's inequality [1, Theorem 9].

Lemma 4.1. *Let $\alpha > 0$. The function, defined on $\mathbb{R}_{>0}^d$,*

$$\vec{x} \mapsto \left(\sum_{i=1}^d x_i^\alpha \right)^{1/\alpha}, \quad (3)$$

1. *is convex if $\alpha > 1$;*
2. *is concave if $0 < \alpha < 1$.*

The displayed power mean is not defined at $\alpha = 0$. Whenever the entropy parameter equals 0, the proof instead uses the endpoint formula for H_0 .

The second fact is the standard convexity criterion for multivariate monomials [3, Lemma 8]; see also Ref. [2, Proposition 13].

Lemma 4.2. *Let $N \in \mathbb{N}$ and let $\vec{\beta} = (\beta_1, \dots, \beta_N) \in \mathbb{R}^N$ satisfy $\sum_{\ell=1}^N \beta_\ell = 1$. The function, defined on $\mathbb{R}_{>0}^N$,*

$$\vec{y} \mapsto \prod_{\ell=1}^N y_\ell^{\beta_\ell}, \quad (4)$$

1. *is concave if and only if $\beta_\ell \geq 0$ for every ℓ ;*
2. *is convex if and only if there is a unique index k such that $\beta_k > 0$ and $\beta_\ell \leq 0$ for every $\ell \neq k$.*

Combining these facts gives the finite-support result.

Lemma 4.3. *Let τ be a discrete probability measure supported on finitely many points $\alpha_1 < \dots < \alpha_N$ of $[0, +\infty]$, and let $t \in \mathbb{R} \setminus \{0\}$. For every $d \in \mathbb{N}$, the function*

$$\vec{x} \mapsto \|\vec{x}\|_1 \exp \left(t \int_{[0, +\infty]} H_\alpha(\hat{x}) d\tau(\alpha) \right) \quad (5)$$

on $\mathbb{R}_{\geq 0}^d \setminus \{0\}$ is

1. *concave if $t > 0$, $\alpha_1, \dots, \alpha_N < 1$, and*

$$\sum_{\ell=1}^N \frac{\alpha_\ell}{1 - \alpha_\ell} \tau(\{\alpha_\ell\}) \leq \frac{1}{t}; \quad (6)$$

2. *convex if $t < 0$ and either*

(a) $\alpha_1, \dots, \alpha_N \leq 1$, or

(b) $\alpha_1, \dots, \alpha_{N-1} < 1$, $\alpha_N > 1$, and

$$\sum_{\ell=1}^N \frac{\alpha_\ell}{1 - \alpha_\ell} \tau(\{\alpha_\ell\}) \leq \frac{1}{t}, \quad (7)$$

where $\alpha/(1 - \alpha)$ has value -1 at $\alpha = +\infty$.

Proof sketch. First assume that every atom satisfies $0 < \alpha_\ell < +\infty$ and $\alpha_\ell \neq 1$, and set

$$\beta_\ell = t \frac{\alpha_\ell}{1 - \alpha_\ell} \tau(\{\alpha_\ell\}), \quad \bar{\beta} = 1 - \sum_{\ell=1}^N \beta_\ell. \quad (8)$$

The function factors as

$$\left(\sum_i x_i \right)^{\bar{\beta}} \prod_{\ell=1}^N \left(\sum_i x_i^{\alpha_\ell} \right)^{\beta_\ell / \alpha_\ell}. \quad (9)$$

Lemma 4.1 supplies the convexity or concavity of each power mean, and Lemma 4.2 supplies the joint sign criterion for the outer monomial. When $t < 0$ and all support lies in $[0, 1]$, no moment bound is needed. Atoms at 0, 1, and $+\infty$ are handled by the endpoint formulas and endpoint-preserving perturbations or pointwise limits; the branch containing $\alpha = 1$ occurs only in the all-lower negative case. $\square$

B. Sufficient conditions for the temperate homomorphisms - general measure

We now extend the result to a general probability measure τ . The signed integral

$$\int_{[0,1]} \frac{\alpha}{1 - \alpha} d\tau(\alpha) + \int_{(1,+\infty]} \frac{\alpha}{1 - \alpha} d\tau(\alpha) \leq \frac{1}{t} \quad (10)$$

is used only when the two one-sided integrals are finite; at $+\infty$ the integrand is -1 . In particular, Eq. (10) never hides an undefined cancellation at $\alpha = 1$. The first integral is the *lower one-sided moment*, and the second is the *upper one-sided moment*; moment-finiteness means that the relevant integral is an ordinary finite real number. We call support in $[0, 1]$ *all-lower support*, and support in $(1, +\infty]$ *upper support*. The exceptional negative branch allows all-lower support together with one isolated point $\alpha_* > 1$. The *Shannon point*, atom, or branch refers to $\alpha = 1$.

Proposition 4.4. *Let τ be a probability measure on $[0, +\infty]$ and let $t \in \mathbb{R} \setminus \{0\}$. The function in Eq. (2) is*

1. *concave if $t > 0$, $\text{supp}(\tau) \subseteq [0, 1]$, $\tau(\{1\}) = 0$, and*

$$\int_{[0,1]} \frac{\alpha}{1 - \alpha} d\tau(\alpha) \leq \frac{1}{t}; \quad (11)$$

2. *convex if $t < 0$ and either*

- (a) $\text{supp}(\tau) \subseteq [0, 1]$, with no moment-finiteness condition, or
- (b) there is $\alpha_* \in (1, +\infty]$ such that $\text{supp}(\tau) \subseteq [0, 1] \cup \{\alpha_*\}$, $\tau(\{1\}) = 0$, the two one-sided integrals in Eq. (10) are finite, and Eq. (10) holds.

Proof sketch. Use one partition of $[0, +\infty]$ for every vector appearing in the convexity inequality and push the mass of each cell to a common representative. On $[0, 1)$ the representatives are chosen from below, so the discrete moment bound is preserved. The Shannon branch is kept separate, and in the exceptional negative case the point α_* is fixed throughout the approximation. Each finite approximation satisfies Lemma 4.3.

Here a *cell* is one set in the finite partition, and its *representative* is the single parameter value to which the mass of that cell is moved. Calling the discretization common means that these cells and representatives do not depend on which vector in the inequality is being evaluated.

For fixed dimension, $0 \leq H_\alpha(\hat{x}) \leq \log d$, including the endpoint values. Dominated convergence therefore gives pointwise convergence of the entropy integrals and of their exponentials, and a pointwise limit preserves the required convexity or concavity inequality. This is the only convergence input needed here. Notice especially that the all-lower branch with $t < 0$ requires no integrability of $\alpha/(1 - \alpha)$ near 1. Here dominated convergence means that one common integrable bound allows the limit to pass through the integral, and pointwise means at each fixed vector $\vec{x}$. $\square$

C. Monotonicity under mixing channels

As mentioned at the beginning of this section, monotonicity under mixing channels acting on the first register X always holds. Mixing on X increases every $H_\alpha(P_X)$ and hence the integrated entropy. Consequently $\Phi_{t,\tau}$ increases for $t > 0$ and decreases for $t < 0$; after applying $(1/t) \log$, the associated conditional entropy increases in both cases. This gives the appropriate positive- and negative-homomorphism order directions. Using Proposition A.1 and Lemma A.3 of Ref. [4], the columnwise concavity or convexity established above is converted into monotonicity under the conditioning operation and then lifted to the original conditionally mixing channels. The derivation case follows from Lemmas A.2 and A.4 of Ref. [4].

D. Sufficient conditions for tropical homomorphisms

We next take the tropical limits. Monotonicity is preserved under pointwise limits, so the temperate conditions immediately provide the admissible endpoint families.

Proposition 4.5. *Let τ be a probability measure on $[0, +\infty]$. The function*

$$P_{XY} \mapsto \min_{\substack{y \in \mathcal{Y} \\ P_Y(y) > 0}} \int_{[0, +\infty]} H_\alpha(P_{X|Y=y}) d\tau(\alpha) \quad (12)$$

is monotone under conditionally mixing channels if either

- (a) $\text{supp}(\tau) \subseteq [0, 1]$, with no moment-finiteness condition, or
- (b) there is $\alpha_* \in (1, +\infty]$ such that $\text{supp}(\tau) \subseteq [0, 1] \cup \{\alpha_*\}$, $\tau(\{1\}) = 0$, both one-sided integrals are finite, and

$$\int_{[0,1]} \frac{\alpha}{1 - \alpha} d\tau(\alpha) + \int_{(1,+\infty]} \frac{\alpha}{1 - \alpha} d\tau(\alpha) \leq 0. \quad (13)$$

Proof sketch. Use

$$\lim_{t \rightarrow -\infty} \frac{1}{t} \log \left(\sum_{y \in \mathcal{Y}} P_Y(y) \exp \left(t \int H_\alpha(P_{X|Y=y}) d\tau(\alpha) \right) \right) \quad (14)$$

$$= \min_{\substack{y \in \mathcal{Y} \\ P_Y(y) > 0}} \int H_\alpha(P_{X|Y=y}) d\tau(\alpha). \quad (15)$$

The all-lower case is the limit of Proposition 4.4(2a). If the signed moment is strictly negative, the exceptional case holds for all sufficiently large $|t|$. At equality, move an arbitrarily small amount of mass to α_* , apply the strict case, and take a final pointwise limit. This retains both normalization and the boundary of the parameter range. $\square$

Proposition 4.6. *The function*

$$P_{XY} \mapsto \max_{\substack{y \in \mathcal{Y} \\ P_Y(y) > 0}} H_0(P_{X|Y=y}) \quad (16)$$

is monotone under conditionally mixing channels.

Proof sketch. This is the $t \rightarrow +\infty$ limit of the positive temperate branch with τ concentrated at $\alpha = 0$:

$$\lim_{t \rightarrow +\infty} \frac{1}{t} \log \left(\sum_{y \in \mathcal{Y}} P_Y(y) e^{tH_0(P_{X|Y=y})} \right) = \max_{\substack{y \in \mathcal{Y} \\ P_Y(y) > 0}} H_0(P_{X|Y=y}). \quad (17)$$

$\square$

Remark 4.7. Unless $\alpha = 0$, the positive temperate moment bound is eventually violated as $t \rightarrow +\infty$. Thus the positive tropical family contains only the $\alpha = 0$ point mass. In contrast, every point mass is allowed in the negative tropical family, subject to the two alternatives in Proposition 4.5.

E. Sufficient conditions for derivations

We now turn to the derivations.

Proposition 4.8. *The function*

$$P_{XY} \mapsto \sum_{y \in \mathcal{Y}} P_Y(y) H_\alpha(P_{X|Y=y}) \quad (18)$$

is monotone under conditionally mixing channels if $\alpha \in [0, 1]$.

Proof sketch. For the point mass at the actual parameter α , take $t \rightarrow 0^-$ in the all-lower branch of Proposition 4.4. The finite log-sum-exp identity

$$\lim_{t \rightarrow 0} \frac{1}{t} \log \left(\sum_{y \in \mathcal{Y}} P_Y(y) e^{tH_\alpha(P_{X|Y=y})} \right) = \sum_{y \in \mathcal{Y}} P_Y(y) H_\alpha(P_{X|Y=y}) \quad (19)$$

passes monotonicity to the limit. Equivalently, this is the concavity of the perspective $\vec{x} \mapsto \|\vec{x}\|_1 H_\alpha(\hat{x})$ used in the derivation lift. The word *perspective* means precisely this positively homogeneous extension from normalized probability vectors to nonzero nonnegative vectors; positive homogeneity is the identity $F(s\vec{x}) = sF(\vec{x})$ for $s > 0$. $\square$

These results identify the admissible homomorphisms and derivations. With the normalization of Definition 6.1 in Ref. [4], they give exactly the candidate families in Eqs. (173)–(176) of Ref. [4].

5. NECESSARY CONDITIONS FOR THE PARAMETERS

In this section, we show that the sufficient conditions on the parameters derived in the previous section are also necessary. No assumption is made that the measure vanishes at a prescribed rate as $\alpha \rightarrow 1$, and no global finiteness condition is placed on the weighted integral. In the positive branch, concavity forces the lower one-sided moment to be finite and bounded. In the exceptional negative branch, convexity forces both one-sided moments to be finite, whereas the all-lower negative branch requires no such finiteness.

Our approach consists in identifying explicit perturbation directions and computing the signs of the second derivatives along them. Outside the ranges of Section 4, the resulting curvature has the wrong sign. Sharp counterexamples require increasingly high dimensions, but the underlying idea is the finite-dimensional multivariate monomial

$$(v_0, \dots, v_d) \mapsto v_0^{\beta_0} v_1^{\beta_1} \dots v_d^{\beta_d}, \quad \beta_0 = 1 - \sum_{i=1}^d \beta_i, \quad (20)$$

whose curvature criterion is Lemma 4.2. In the discrete entropy expression, the correspondence is

$$\begin{aligned} v_0 &\longleftrightarrow \sum_j x_j, \\ \beta_i &\longleftrightarrow \tau(\{\alpha_i\}) t \frac{\alpha_i}{1 - \alpha_i}, \\ v_i &\longleftrightarrow \left(\sum_j x_j^{\alpha_i} \right)^{1/\alpha_i}. \end{aligned} \quad (21)$$

The latter quantities are coupled through $\vec{x}$. Blocks whose cardinalities and weights scale with a dimension parameter make different blocks dominate on different ranges of α . For example, appending d entries of weight $1/d$ to a fixed vector $\vec{x} = (x_1, \dots, x_k)$ gives

$$\vec{y} = \left(x_1, \dots, x_k, \underbrace{1/d, \dots, 1/d}_{d \text{ entries}} \right)$$

and hence

$$\left(\sum_j y_j^\alpha \right)^{1/\alpha} = \left(\sum_{j=1}^k x_j^\alpha + d^{1-\alpha} \right)^{1/\alpha}, \quad (22)$$

which diverges for $\alpha < 1$ and converges to $(\sum_{j=1}^k x_j^\alpha)^{1/\alpha}$ for $\alpha > 1$.

There is one analytic point that is important even in this abbreviated account. For each localized path used below, its support is fixed near the perturbation point and its set of maximizing coordinates remains fixed there. On these paths the endpoint-aware kernels and their first two path derivatives extend continuously through 0, 1, and $+\infty$; in particular, the apparent pole at 1 is removable. One differentiates the finite-dimensional kernels before integrating, then truncates

the high-dimensional analysis on the two sides of 1 and passes to the endpoint only afterward. Domination by total variation and monotone convergence make the required moment bounds conclusions of the argument. Thus no derivative is exchanged with an already limiting piecewise envelope, and no separate finiteness assumption is needed.

An *endpoint-aware kernel* is the function $\alpha \mapsto H_\alpha(\widehat{\vec{x} + \lambda \vec{v}})$ with the endpoint values at 0, 1, and $+\infty$ retained. A *removable pole* means that the apparent denominator $1 - \alpha$ is cancelled by a numerator that also vanishes at $\alpha = 1$. *Endpoint-separated truncation* means integrating on closed regions below and above 1 before moving their boundaries towards 1. Domination by total variation means controlling the absolute value of the kernel by an integrable bound for the positive measure $|\nu|$ defined next.

For any finite signed measure ν , meaning a finite measure that may take both positive and negative values, its *total-variation measure* $|\nu|$ is the positive measure that records the absolute mass of ν , and we set $\text{supp}(\nu) := \text{supp}(|\nu|)$. A point s is $|\nu|$ -null when $|\nu|(\{s\}) = 0$; for a positive measure this simply means $\nu(\{s\}) = 0$. Thus a *null point* may still lie in the support: it has no atom at that single point. A *null threshold* is a null point used as the boundary between localization intervals. A finite measure has at most countably many atoms, so such thresholds can be chosen between distinct parameter values.

a. Explicit witness vectors. All counterexamples below are restrictions to affine lines of the form $\lambda \mapsto \vec{x} + \lambda \vec{v}$. Here $\lambda \in \mathbb{R}$ is taken small enough that all coordinates remain positive; $\vec{x}$ is the initial point at $\lambda = 0$, and $\vec{v}$ is its perturbation direction. The curvature of a path is its second derivative at $\lambda = 0$, while a logarithmic derivative is a derivative of the logarithm of the positive function being tested.

Here $d \geq 2$ is an auxiliary integer scale parameter tending to infinity. It is not the actual vector dimension, which is the sum of the block cardinalities $n_{j,d}$. A *block* is a group of repeated equal coordinates, its cardinality is the number of coordinates in that group, and $\lceil s \rceil$ is the least integer not smaller than s . A block velocity is the common relative first-order change assigned to one block. A block *dominates* at a parameter α when its contribution to $\sum_i x_i^\alpha$ has the largest power of d . A *localizer* is a block family arranged so that the dominant block changes with α and thereby isolates the measure on selected parameter intervals. A *witness pair* is the displayed initial point together with its direction.

We record every coordinate inside parentheses. For the two-block family, fix $r > 0$ with $r \neq 1$. The number r is the threshold at which the dominant block changes, and $R := r + 1$ is an auxiliary size exponent. Put

$$n_{0,d} = \lceil d^R \rceil, \quad n_{1,d} = \lceil d^{R-r} \rceil. \quad (23)$$

The initial point and the direction with block velocities (u_0, u_1) are

$$\vec{x}_r^{(d)} = \left(\underbrace{1, \dots, 1}_{n_{0,d}}, \underbrace{d, \dots, d}_{n_{1,d}} \right), \quad (24)$$

$$\vec{v}_r^{(d)}(u_0, u_1) = \left(\underbrace{u_0, \dots, u_0}_{n_{0,d}}, \underbrace{du_1, \dots, du_1}_{n_{1,d}} \right). \quad (25)$$

For the three-block family, fix $0 < a < b < +\infty$, with $a, b \neq 1$. The two numbers a and b are the dominant-block crossing thresholds, and $R := a + b + 1$. Put

$$n_{0,d} = \lceil d^R \rceil, \quad n_{1,d} = \lceil d^{R-a} \rceil, \quad n_{2,d} = \lceil d^{R-a-b} \rceil. \quad (26)$$

Its initial point and direction are

$$\vec{x}_{a,b}^{(d)} = \left(\underbrace{1, \dots, 1}_{n_{0,d}}, \underbrace{d, \dots, d}_{n_{1,d}}, \underbrace{d^2, \dots, d^2}_{n_{2,d}} \right), \quad (27)$$

$$\vec{v}_{a,b}^{(d)}(u_0, u_1, u_2) = \left(\underbrace{u_0, \dots, u_0}_{n_{0,d}}, \underbrace{du_1, \dots, du_1}_{n_{1,d}}, \underbrace{d^2u_2, \dots, d^2u_2}_{n_{2,d}} \right). \quad (28)$$

Finally, fix a cutoff $c > 1$. It separates the middle parameter range $1 < \alpha < c$ from the upper tail $\alpha > c$ and is not a universal constant. In each use, c is chosen null for the relevant measure, so no atom lies on this boundary. The first two blocks cross at the Shannon point 1, and the second and third cross at c . For the rounded Shannon family put $p = q := 1/2$, $R := c + 2$, and

$$n_{0,d} = \lceil d^R \rceil, \quad n_{1,d} = \lceil d^{R-1} \rceil, \quad n_{2,d} = \lceil d^{R-1-c} \rceil. \quad (29)$$

Its initial point and direction are

$$\vec{x}_{\text{Sh}}^{(d)} = \left(\underbrace{q, \dots, q}_{n_{0,d}}, \underbrace{dp, \dots, dp}_{n_{1,d}}, \underbrace{d^2, \dots, d^2}_{n_{2,d}} \right), \quad (30)$$

$$\vec{v}_{\text{Sh}}^{(d)}(h, v) = \left(\underbrace{h/\log d, \dots, h/\log d}_{n_{0,d}}, \underbrace{-dh/\log d, \dots, -dh/\log d}_{n_{1,d}}, \underbrace{d^2v, \dots, d^2v}_{n_{2,d}} \right). \quad (31)$$

In the first two families $u_0, u_1, u_2 \in \mathbb{R}$ are the block velocities. In the Shannon family, $p = q = 1/2$ are fixed scalar block weights, while $h, v \in \mathbb{R}$ control the first-two-block Shannon perturbation and the third block, respectively; the scalar v is unrelated to a direction vector $\vec{v}$. The subscript Sh stands for Shannon, and “rounded” refers to the ceiling signs that turn the desired block sizes into integers. The velocities will be specified separately for each obstruction.

A. Necessary conditions; temperate case

The temperate case corresponds to the bulk entropies $H_{t,\tau}$ among the candidate families of Ref. [4]; “bulk” here means the finite-temperature, or temperate, family. As above, it is sufficient to study Eq. (2). For notational convenience, combine the weight t with the probability measure τ and write

$$\mu := t\tau. \quad (32)$$

A measure μ is positive when $\mu(E) \geq 0$ for every measurable set E , and negative when $\mu(E) \leq 0$ for every such E . Since $\mu = t\tau$ is one-signed, $\mu > 0$ means $t > 0$, $\mu < 0$ means $t < 0$, and $|\mu| = |t|\tau$. As above, $\text{supp}(\mu)$ means the support of $|\mu|$. A lower or upper *truncated coefficient* is a weighted integral over $[0, a)$ or $(b, +\infty]$, with $a < 1 < b$; it therefore stays away from the Shannon point until the two limits are taken separately. Whenever the two one-sided integrals are finite, we also have the coefficient

$$\beta_0 = 1 - \int_{[0, +\infty] \setminus \{1\}} \frac{\alpha}{1 - \alpha} d\mu(\alpha), \quad (33)$$

where the integral is the sum of its parts on $[0, 1)$ and $(1, +\infty]$.

Whenever a signed moment is written as a single ordinary real number below, it is the sum of the two one-sided integrals after their finiteness has been established. Finiteness is not a blanket hypothesis on μ : before it is known, the argument works with endpoint-separated truncations. The localized estimates show that if convexity leaves any support above 1, the relevant one-sided moments must be finite. If all support lies in $[0, 1]$, no such condition is imposed.

We first treat $\mu > 0$, and then $\mu < 0$. In the negative case the Shannon atom, the signed moment, and the number of upper support points are isolated in separate steps, exactly matching the alternatives in Proposition 4.4.

Proposition 5.1. *Let μ be a positive finite measure on $[0, +\infty]$. The function*

$$\vec{x} \mapsto \|\vec{x}\|_1 \exp\left(\int_{[0, +\infty]} H_\alpha(\hat{x}) d\mu(\alpha)\right) \quad (34)$$

is concave only if $\text{supp}(\mu) \subseteq [0, 1]$, $\mu(\{1\}) = 0$, and

$$\int_{[0, 1)} \frac{\alpha}{1 - \alpha} d\mu(\alpha) \leq 1. \quad (35)$$

Proof sketch. The three exclusions use the finite-dimensional witness families, or localizers, displayed above. An atom $m = \mu(\{1\}) > 0$ is detected by choosing a μ -null point $c > 1$ and using the complete point-direction pair

$$\left(\vec{x}_{\text{Sh}}^{(d)}, \vec{v}_{\text{Sh}}^{(d)}(1, 0)\right). \quad (36)$$

Its limiting logarithmic derivatives are m and 0, so the outer exponential has limiting curvature $m^2 > 0$.

For the remaining two tests, choose a μ -null threshold $r > 0$, put $R = r + 1$, and use the two-block initial point in Eq. (24). If $r > 1$ and upper support is present, the complete pair is

$$\left(\vec{x}_r^{(d)}, \vec{v}_r^{(d)}(0, 1)\right), \quad (37)$$

and the localized coefficient

$$A_r = \int_{(r, +\infty]} \frac{\alpha}{1 - \alpha} d\mu(\alpha)$$

is negative. The limiting first and second logarithmic derivatives are A_r and $-A_r$, hence the curvature is $A_r^2 - A_r > 0$. Thus no upper support is possible. If $r < 1$, use instead the complete pair

$$\left(\vec{x}_r^{(d)}, \vec{v}_r^{(d)}(1, 0)\right) \quad (38)$$

and write

$$B_r = \int_{[0, r)} \frac{\alpha}{1 - \alpha} d\mu(\alpha).$$

The limiting derivatives are B_r and $-B_r$, so concavity implies $B_r(B_r - 1) \leq 0$ and therefore $B_r \leq 1$. Letting null thresholds increase to 1 and using monotone convergence proves the stated moment bound. Endpoint-separated truncation justifies the limits without a preliminary finiteness hypothesis. Monotone convergence here means that the increasing nonnegative truncated integrands may be passed to their limiting integral. $\square$

Next, let μ be negative.

Proposition 5.2. *Let μ be a negative finite measure with $\mu(\{1\}) \neq 0$. Then the function*

$$\vec{x} \mapsto \|\vec{x}\|_1 \exp\left(\int_{[0,+\infty]} H_\alpha(\hat{x}) d\mu(\alpha)\right) \quad (39)$$

is convex only if $\text{supp}(\mu) \subseteq [0, 1]$.

Proof sketch. Assume that both the Shannon atom and upper support are present. If $m := \mu(\{1\}) < 0$, choose a $|\mu|$ -null point $c > 1$ such that $\mu((c, +\infty]) < 0$, and define the localized upper contribution

$$A_c := \int_{(c, +\infty]} \frac{\alpha}{1 - \alpha} d\mu(\alpha) > 0.$$

Use

$$\left(\vec{x}_{\text{Sh}}^{(d)}, \vec{v}_{\text{Sh}}^{(d)}\left(-\frac{A_c}{m}, 1\right)\right). \quad (40)$$

The first logarithmic derivative then tends to zero and the second tends to $-A_c < 0$. This is a genuinely concave direction and contradicts convexity. Working first on truncated ranges makes the calculation finite; the endpoint limits are taken only after the curvature sign is fixed. $\square$

Proposition 5.3. *Let μ be a negative finite measure with $\mu(\{1\}) = 0$, and suppose that the function*

$$\vec{x} \mapsto \|\vec{x}\|_1 \exp\left(\int_{[0,+\infty]} H_\alpha(\hat{x}) d\mu(\alpha)\right) \quad (41)$$

is convex. Let $0 < a < 1 < b < +\infty$ be $|\mu|$ -null thresholds with $\mu((b, +\infty]) < 0$, and set

$$B_a = \int_{[0,a)} \frac{\alpha}{1 - \alpha} d\mu(\alpha) \leq 0, \quad A_b = \int_{(b,+\infty]} \frac{\alpha}{1 - \alpha} d\mu(\alpha) > 0. \quad (42)$$

Thus B_a is the lower truncated coefficient and A_b the upper truncated coefficient. Then

$$A_b + B_a \geq 1. \quad (43)$$

Proof sketch. If Eq. (43) failed, then $C := 1 - A_b - B_a > 0$. This C is the remaining coefficient of the middle block, which contains $\alpha = 1$. Use the three-block pair

$$\left(\vec{x}_{a,b}^{(d)}, \vec{v}_{a,b}^{(d)}\left(0, \frac{1}{C}, -\frac{1}{A_b}\right)\right), \quad (44)$$

where $R = a + b + 1$. The velocity

$$(u_0, u_1, u_2) = \left(0, \frac{1}{C}, -\frac{1}{A_b}\right)$$

has limiting first logarithmic derivative 0 and limiting second derivative

$$-\frac{1}{C} - \frac{1}{A_b} < 0.$$

This gives a finite-dimensional violation of convexity. The statement is made only with truncated coefficients: no singular signed moment is formed before its two one-sided parts are known to be finite. $\square$

Remark 5.4. The points used in these counterexamples need not initially be normalized. The function is positively homogeneous on the whole semiring domain, so rescaling by a positive constant does not change the curvature sign. Here normalized means that the coordinates sum to one, and positively homogeneous means $F(s\vec{x}) = sF(\vec{x})$ for $s > 0$. Equivalently, the counterexample can be embedded into a normalized joint probability matrix, interpreted in the conditional-majorization semiring of Ref. [4], and lifted using Proposition A.1 and Lemma A.3 of Ref. [4].

Remark 5.5. A nontraceless direction can likewise be embedded into a trace-preserving two-column perturbation. A direction is traceless when $\sum_i v_i = 0$; trace-preserving means that the total mass of all columns remains constant. Let $\vec{x}' > 0$ be a second-column base point and put $\hat{x}' := \vec{x}' / \|\vec{x}'\|_1$. If the first column changes as $\vec{x} + \lambda\vec{v}$, change the second as

$$\vec{x}' - \lambda\hat{x}' \sum_i v_i. \quad (45)$$

The opposite sign is essential: it keeps the total weight fixed. Additivity over the conditioning register makes the second column affine in λ , so it does not alter the curvature obstruction.

Finally, the upper support cannot contain two distinct points.

Proposition 5.6. *Let μ be a negative finite measure with $\mu(\{1\}) = 0$, and suppose that the function*

$$\vec{x} \mapsto \|\vec{x}\|_1 \exp\left(\int_{[0,+\infty]} H_\alpha(\hat{x}) d\mu(\alpha)\right) \quad (46)$$

is convex. Then either $\text{supp}(\mu) \subseteq [0, 1]$, or there is a unique $\alpha_ \in (1, +\infty]$ such that*

$$\text{supp}(\mu) \subseteq [0, 1] \cup \{\alpha_*\}. \quad (47)$$

In the second case the two one-sided moments are finite and satisfy

$$\int_{[0,1]} \frac{\alpha}{1-\alpha} d\mu(\alpha) + \int_{(1,+\infty]} \frac{\alpha}{1-\alpha} d\mu(\alpha) \geq 1. \quad (48)$$

Proof sketch. Suppose that $1 < \alpha_1 < \alpha_2 \leq +\infty$ are two support points. Choose $|\mu|$ -null thresholds $1 < a < \alpha_1 < b < \alpha_2$; when $\alpha_2 = +\infty$, take any finite null $b > \alpha_1$ that separates the two support portions. Define

$$A_1 := \int_{(a,b)} \frac{\alpha}{1-\alpha} d\mu(\alpha) > 0, \quad A_2 := \int_{(b,+\infty]} \frac{\alpha}{1-\alpha} d\mu(\alpha) > 0.$$

These are positive because both μ and $\alpha/(1-\alpha)$ are negative above 1. The two intervals are localized in different dominant blocks. The complete witness pair is

$$\left(\vec{x}_{a,b}^{(d)}, \vec{v}_{a,b}^{(d)}\left(0, \frac{1}{A_1}, -\frac{1}{A_2}\right)\right). \quad (49)$$

Its direction cancels the limiting first derivative, while the limiting second derivative is

$$-\frac{1}{A_1} - \frac{1}{A_2} < 0. \quad (50)$$

This contradicts convexity. Shrinking the two localization windows shows that the upper support has at most one point. The finiteness and signed inequality in the exceptional branch now follow from Proposition 5.3: the single upper atom gives a finite upper moment, its truncated inequality bounds the lower terms, and monotone convergence gives lower-moment finiteness. Only then are the two one-sided integrals combined. $\square$

Thus the last two propositions cover all negative measures without a global finiteness assumption; the all-lower branch still carries no moment condition.

B. Necessary conditions for tropical homomorphisms

The tropical counterexamples use the same finite witness families as the temperate proof, but quasiconvexity permits a second-derivative test only after the first derivative has been made exactly zero at the finite dimension under consideration. Here quasiconvexity means

$$G(\lambda x + (1 - \lambda)z) \leq \max\{G(x), G(z)\}, \quad 0 \leq \lambda \leq 1.$$

A direction is *stationary* when its first derivative is zero. A *transverse direction* is a correction direction whose first derivative is nonzero, so a small multiple of it can impose exact stationarity.

Proposition 5.7. *Let τ be a probability measure on $[0, +\infty]$. If the function in Eq. (12) is monotone under conditionally mixing channels, then either*

(a) $\text{supp}(\tau) \subseteq [0, 1]$; or

(b) *there is a unique $\alpha_* \in (1, +\infty]$ such that*

$$\text{supp}(\tau) \subseteq [0, 1] \cup \{\alpha_*\}, \quad \tau(\{1\}) = 0,$$

both one-sided integrals

$$\int_{[0,1]} \frac{\alpha}{1-\alpha} d\tau(\alpha), \quad \int_{(1,+\infty]} \frac{-\alpha}{1-\alpha} d\tau(\alpha)$$

are finite, and

$$\int_{[0,+\infty] \setminus \{1\}} \frac{\alpha}{1-\alpha} d\tau(\alpha) \leq 0. \quad (51)$$

Proof sketch. A Shannon localizer first excludes $\tau(\{1\}) > 0$ whenever upper support is present. Set $\mu := -\tau$. Then the signed Shannon atom is $m := \mu(\{1\}) = -\tau(\{1\}) < 0$. Choose a $|\mu|$ -null point $c > 1$ with $\mu((c, +\infty]) < 0$ and define

$$A_c := \int_{(c,+\infty]} \frac{\alpha}{1-\alpha} d\mu(\alpha) > 0.$$

Use the initial point and the two displayed directions

$$\left(\vec{x}_{\text{Sh}}^{(d)}, \vec{v}_{\text{Sh}}^{(d)} \left(-\frac{A_c}{m}, 1 \right) \right), \quad \left(\vec{x}_{\text{Sh}}^{(d)}, \vec{v}_{\text{Sh}}^{(d)}(0, 1) \right), \quad (52)$$

corresponding to the block-velocity vectors $u = (-A_c/m, 1)$ and $e = (0, 1)$. The limiting first derivative along u is zero, the second derivative is $-A_c < 0$, and the limiting first derivative along e is $A_c \neq 0$.

If two separated upper-support portions remain, choose support points $1 < \alpha_1 < \alpha_2 \leq +\infty$ and $|\mu|$ -null thresholds $1 < a < \alpha_1 < b < \alpha_2$, with the same finite- b interpretation as above when $\alpha_2 = +\infty$. Define

$$A_1 := \int_{(a,b)} \frac{\alpha}{1-\alpha} d\mu(\alpha) > 0, \quad A_2 := \int_{(b,+\infty]} \frac{\alpha}{1-\alpha} d\mu(\alpha) > 0.$$

These coefficients give the initial point and directions

$$\left(\vec{x}_{a,b}^{(d)}, \vec{v}_{a,b}^{(d)} \left(0, \frac{1}{A_1}, -\frac{1}{A_2} \right) \right), \quad \left(\vec{x}_{a,b}^{(d)}, \vec{v}_{a,b}^{(d)}(0, 1, 0) \right), \quad (53)$$

corresponding to $u = (0, 1/A_1, -1/A_2)$ and $e = (0, 1, 0)$. The limiting first derivative along u is zero, its second derivative is $-1/A_1 - 1/A_2 < 0$, and the first derivative along e tends to $A_1 \neq 0$. If Eq. (51) fails in the exceptional branch, choose $|\mu|$ -null thresholds $0 < a < 1 < b < \alpha_*$ close enough to 1 that

$$B_a := \int_{[0,a]} \frac{\alpha}{1-\alpha} d\mu(\alpha) \leq 0, \quad A_b := \int_{(b,+\infty]} \frac{\alpha}{1-\alpha} d\mu(\alpha) > 0$$

satisfy $A_b + B_a < 0$, and define

$$C := -A_b - B_a > 0.$$

This tropical coefficient has no added 1: unlike the temperate function, the tropical one-column function has no outer $\|\vec{x}\|_1$ term. The coefficients $A_b, C > 0$ give

$$\left(\vec{x}_{a,b}^{(d)}, \vec{v}_{a,b}^{(d)} \left(0, \frac{1}{C}, -\frac{1}{A_b} \right) \right), \quad \left(\vec{x}_{a,b}^{(d)}, \vec{v}_{a,b}^{(d)}(0, 0, 1) \right), \quad (54)$$

corresponding to $u = (0, 1/C, -1/A_b)$ and $e = (0, 0, 1)$. Here the limiting first derivative along u is zero, its second derivative is $-1/C - 1/A_b < 0$, and the first derivative along e tends to $A_b \neq 0$.

For each of the three witnesses, let u and e have the values displayed in the relevant formula. At every sufficiently large finite dimension, replace u by $u + \theta_d e$, where $\theta_d \in \mathbb{R}$ is the scalar correction chosen so that the first derivative is exactly zero. The transverse first derivative tends to a nonzero number, whereas the first derivative along u tends to zero, so $\theta_d \rightarrow 0$ and the corrected second derivative remains strictly negative. Let $g_d(\lambda)$ be the negative tropical one-column function restricted to this corrected affine line. Then

$$g_d'(0) = 0, \quad g_d''(0) < 0. \quad (55)$$

Taylor expansion gives, for every sufficiently small $\varepsilon > 0$,

$$g_d(-\varepsilon) < g_d(0), \quad g_d(\varepsilon) < g_d(0). \quad (56)$$

Quasiconvexity at the midpoint would instead require

$$g_d(0) \leq \max\{g_d(-\varepsilon), g_d(\varepsilon)\}, \quad (57)$$

which is impossible. Thus all three negative second derivatives give genuine finite-dimensional tropical counterexamples. Support isolation makes the upper moment finite; monotone passage through lower truncations then proves lower-moment finiteness. $\square$

Proposition 5.8. *Let τ be a probability measure on $[0, +\infty]$. If*

$$P_{XY} \mapsto \max_{\substack{y \in \mathcal{Y} \\ P_Y(y) > 0}} \int_{[0,+\infty]} H_\alpha(P_{X|Y=y}) d\tau(\alpha) \quad (58)$$

is monotone under conditionally mixing channels, then $\tau = \delta_0$. Consequently, Eq. (58) is precisely the function in Proposition 4.6.

Proof sketch. For a finite probability vector s , write

$$A_\tau(s) = \int_{[0,+\infty]} H_\alpha(s) d\tau(\alpha). \quad (59)$$

For probability vectors p, q of the same dimension, positive-tropical monotonicity gives the max-quasiconcavity inequality

$$A_\tau(\lambda p + (1 - \lambda)q) \geq \max\{A_\tau(p), A_\tau(q)\}, \quad 0 < \lambda < 1. \quad (60)$$

A fixed-support simplex face is the set of probability vectors with one fixed pattern of allowed nonzero coordinates; its relative interior consists of the vectors strictly positive on every allowed coordinate. Applying the displayed inequality in both directions on such a face makes A_τ constant on its relative interior. Apply it to the binary points

$$p = \left(\frac{1}{3}, \frac{2}{3}\right), \quad u = \left(\frac{1}{2}, \frac{1}{2}\right).$$

Here p is a probability vector, unrelated to the scalar Shannon block weight $p = 1/2$ used above, and u is the uniform binary vector. They have the same support, so $A_\tau(p) = A_\tau(u) = \log 2$. Moreover, $H_0(p) = \log 2$ and $H_\alpha(p) < \log 2 = H_\alpha(u)$ for every $\alpha \in (0, +\infty]$. Hence

$$0 = \int_{[0, +\infty]} (\log 2 - H_\alpha(p)) \, d\tau(\alpha). \quad (61)$$

The integrand is nonnegative everywhere and strictly positive on $(0, +\infty]$, so $\tau((0, +\infty]) = 0$ and $\tau = \delta_0$. $\square$

C. Necessary conditions for derivations

Next, we turn to the necessary conditions for the extremal derivations. Under Eq. (78) and Proposition 3.7 of Ref. [4], these are

$$H_{0,\alpha}(P_{XY}) = \sum_{y \in \mathcal{Y}} P_Y(y) H_\alpha(P_{X|Y=y}). \quad (62)$$

It is enough to work directly at the measure level. Thus consider

$$\vec{x} \mapsto \|\vec{x}\|_1 \int_{[0, +\infty]} H_\alpha(\hat{x}) \, d\tau(\alpha), \quad (63)$$

for a finite positive measure τ . Point masses recover the extremal derivations used in Lemmas A.2 and A.4 of Ref. [4]. Here *extremal* means the point-mass case $\tau = \delta_\alpha$; a general represented derivation is a positive integral mixture of these cases.

Proposition 5.9. *Let τ be a finite positive measure. The function*

$$\vec{x} \mapsto \|\vec{x}\|_1 \int_{[0, +\infty]} H_\alpha(\hat{x}) \, d\tau(\alpha)$$

is concave only if $\text{supp}(\tau) \subseteq [0, 1]$. In particular, $\vec{x} \mapsto \|\vec{x}\|_1 H_\alpha(\hat{x})$ is concave only if $\alpha \in [0, 1]$.

Proof sketch. If the support meets $(1, +\infty]$, choose a τ -null threshold $r > 1$ with $\tau((r, +\infty]) > 0$, put $R = r + 1$, and define afresh

$$A_r := \int_{(r, +\infty]} \frac{\alpha}{1 - \alpha} \, d\tau(\alpha) < 0.$$

Use the complete pair

$$\left(\vec{x}_r^{(d)}, \vec{v}_r^{(d)}(0, 1)\right). \quad (64)$$

Its localized upper coefficient A_r is negative, while the second derivative of the normalized derivation converges to $-A_r > 0$. This is a finite-dimensional convex direction and contradicts concavity, including when the upper support point is $+\infty$. $\square$

Combining Eq. (78) and Proposition 3.7 of Ref. [4] with Proposition 4.8 and the semiring lift, gives the final statement.

For clarity, S is the conditional-majorization semiring of Ref. [4]; its elements are represented by unnormalized joint probability matrices, and its addition and multiplication are direct sum and tensor product. The distinguished norm $\|P\|$ is the total weight, namely the sum of all entries of P . A derivation Δ at this norm is additive and satisfies

$$\Delta(P \otimes Q) = \Delta(P)\|Q\| + \|P\|\Delta(Q).$$

Proposition 5.10. *The set of derivations of the conditional majorization semiring S at $\|\cdot\|$ is spanned by the quantities $H_{0,\alpha}$ of Definition 6.1 in Ref. [4], with $\alpha \in [0, 1]$.*

The span assertion in Proposition 5.10 is imported from Eq. (78) and Proposition 3.7 of Ref. [4]. Here “spanned” means generated by positive barycentres: a represented derivation has the form $\int H_{0,\alpha} d\tau(\alpha)$ for a finite positive measure τ ; under the normalization of Definition 6.1 in Ref. [4], τ is a probability measure. The sufficiency and curvature arguments in this section supply the exact restriction $\alpha \in [0, 1]$ for its extremal candidates.

Together, the temperate, tropical, and derivation statements give the exact parameter ranges for the candidate families in Eqs. (173)–(176) of Ref. [4]. Together with the preceding representation results, this completes the parameter classification.

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